BIAS SIMILARITY MEASUREMENT: A BLACK-BOX AUDIT OF FAIRNESS ACROSS LLMS

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ABSTRACT

Fairness audits for large language models are usually reported as per-model scores, which makes it difficult to answer practical questions about lineage, regression, and procurement: do two models fail in the same way, does instruction tuning genuinely debias a family, and are low bias scores produced by better representations or by stronger refusal policies? We present Bias Similarity Measurement (BSM), a black-box framework that treats fairness as a relational property and compares models through coordinated scalar, distributional, behavioral, and representational views. BSM combines disambiguated accuracy, benchmark-native bias scores, output histograms, cosine and Jensen-Shannon distributional distances, ambiguous-case flip behavior into `Unknown`, and centered kernel alignment for open-weight base–tuned pairs. We instantiate the framework using archived evaluations over BBQ, UnQover, and an open-ended StereoSet pipeline, with the main quantitative study centered on the released BBQ and CKA artifacts. Across the archived runs, instruction tuning often suppresses visibly biased behavior by increasing abstention on ambiguous BBQ items, while hidden-state geometry remains highly stable: average diagonal CKA stays between 0.941 and 0.991 and full-matrix CKA between 0.851 and 0.911. The same audit also reveals non-monotonic behavior: LLaMA 3.1 8B becomes markedly more neutral after tuning, whereas LLaMA 3.2 3B becomes more stereotypical under answer-required settings. These results argue for comparative fairness auditing: models that look similarly fair under one benchmark can have materially different bias signatures once refusal, forced choice, and representation are analyzed jointly.

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1 INTRODUCTION

Large language models now appear in procurement pipelines, release gates, and regression dashboards, yet fairness evaluation is still usually framed as a set of isolated leaderboard entries. That framing is incomplete. A deployment team rarely asks only whether one model has a lower bias score than another; it also asks whether a new release inherits the same failure modes as its predecessor, whether a tuned model is safer because it learned different internal associations or because it refuses to answer, and whether an open-weight substitute behaves similarly enough to a proprietary option to be considered an operational replacement. These are fundamentally comparative questions.

This paper introduces *Bias Similarity Measurement* (BSM), a black-box audit framework for comparing fairness signatures across models rather than grading each model independently. BSM is motivated by recent work showing that LLM bias is heterogeneous across tasks, prompting strategies, and social categories (Ferrara, 2023; Wang et al., 2025; Kumar et al., 2025; Yeh et al., 2023). Standard benchmarks such as BBQ (Parrish et al., 2022), UnQover (Li et al., 2020), and StereoSet (Nadeem et al., 2021) remain indispensable, but they are usually consumed as standalone scores. BSM reorganizes these signals into a multi-view similarity audit that supports model-family comparisons, base–tuned regression analysis, and open-versus-closed model screening.

Our central design choice is to keep the audit black-box wherever possible, then add representation analysis only when weights are available. In practice, BSM separates ambiguous contexts, where abstention may be appropriate, from disambiguated or forced-choice settings, where the model must commit. This distinction matters. The archived experiments show that several tuned models reduce benchmark-visible bias primarily by redirecting ambiguous answers to `Unknown`, while their representational geometry remains largely unchanged. The same audit also finds that instruction

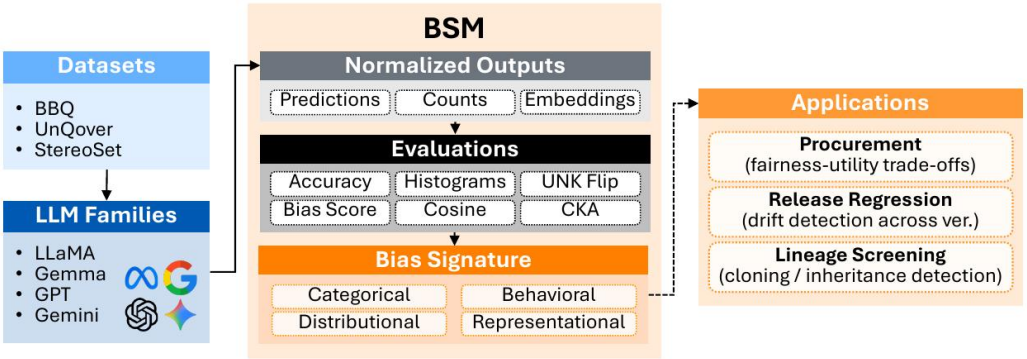

Figure 1: **BSM pipeline.** The audit extracts complementary views of fairness from black-box model behavior and, when available, from base–tuned representation similarity. Ambiguous and answer-required settings are analyzed separately so that refusal-heavy behavior is not conflated with genuine debiasing.

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tuning is not uniformly beneficial: some small models become *more* stereotypical once abstention is removed from the decision space.

The present writeup is intentionally tied to the released artifacts. The repository contains an empty paper template, three summary tables, four finalized figures, and analysis notebooks with archived prediction files and visualization code. We therefore formulate BSM as a method paper grounded in those experiment traces rather than in the preliminary hypothesis document. Concretely, the main quantitative evidence comes from: (i) a 30-endpoint BBQ accuracy sweep over nine social dimensions, corresponding to roughly 0.88 million model responses; (ii) a 29-endpoint four-dimension BBQ audit used for bias-score, histogram, and flip analysis; (iii) UnQover distributional analyses in the notebooks; and (iv) CKA comparisons for five open-weight base–tuned families.

Our contributions are:

- We introduce BSM, a relational fairness audit that compares models using coordinated scalar, distributional, behavioral, and representational views instead of a single standalone score.
- We instantiate BSM on archived evaluations across contemporary LLaMA, Gemma, Gemini, GPT, and legacy instruction-tuned baselines, showing how the framework exposes inheritance, convergence, and regressions that are difficult to see in isolated benchmark reports.
- We provide evidence that apparent fairness gains from instruction tuning often come from abstention on ambiguous prompts, while internal geometry stays highly stable under CKA.
- We show that the resulting audit is immediately useful for procurement and release analysis: open-weight tuned Gemma models approach the neutrality of proprietary systems on the reported BBQ bias scores, but small-model regressions and forced-choice behavior still separate model families.

2 Related Work

2.1 Bias Benchmarks for LLMs

Fairness and bias evaluation for LLMs spans surveys, diagnostic toolkits, and compositional benchmarks (Ferrara, 2023; Wang et al., 2025; Kumar et al., 2024; 2025). This literature has established two relevant facts. First, social bias is strongly benchmark-dependent: different tasks surface different forms of harm. Second, a single metric is rarely sufficient, because benchmark format can reward refusal, verbosity, or surface-level alignment rather than changes in underlying associations.

Our work builds directly on three benchmark families. BBQ (Parrish et al., 2022) separates ambiguous contexts, where stereotypes can dominate under weak evidence, from disambiguated contexts, where a model should answer correctly even when the correct answer conflicts with a stereotype. UnQover

(Li et al., 2020) removes that abstention path and probes directional preferences under underspecified questions. StereoSet (Nadeem et al., 2021) targets stereotypical bias in language model completions. BSM does not replace these resources; it reinterprets them as complementary views inside a single comparative audit.

2.2 Why Relational Audits Are Still Missing

Most fairness studies still summarize each model independently. That is useful for benchmarking but weak for operational questions such as regression testing or lineage screening. A tuned model may have a lower benchmark-visible bias score while remaining highly similar to its base model in output distributions or representations. Conversely, two systems with similar scalar scores can reach them through very different behaviors, for example by abstaining more often versus by changing the direction of non-abstaining answers.

The closest prior ideas come from broader comparative evaluation and interfaced LLM auditing (Yeh et al., 2023; Liusie et al., 2024). BSM differs in purpose and design: it focuses specifically on fairness, keeps the core analysis black-box, and organizes multiple benchmark-native signals into a pairwise audit space rather than relying on a single judge or a single score.

2.3 Representational Similarity

Representational similarity methods have long been used to compare neural networks, with CKA emerging as a robust choice for cross-model layer alignment (Kornblith et al., 2019). More recent work has explored representational similarity for LLMs directly (Klabunde et al., 2023). We use this line of work in a deliberately narrow way. CKA is not our fairness metric; it is one view in BSM that helps answer whether observed behavioral changes are accompanied by meaningful internal reorganization. The archived CKA results show that, for several open-weight base–tuned pairs, they often are not.

3 Bias Similarity Measurement

3.1 Audit Objects

BSM treats each model as a bundle of fairness signals collected across benchmarks and prompt regimes. For a model m, we construct a multi-view signature

$$\Phi(m) = \{\phi_{\text{acc}}(m), \phi_{\text{bias}}(m), \phi_{\text{hist}}(m), \phi_{\text{flip}}(m), \phi_{\text{repr}}(m)\},$$

where each view is computed from observable outputs, except ϕ_{repr}, which is included only when hidden states are accessible. For a pair of models (m_i, m_j), BSM compares the corresponding views with view-appropriate similarity measures. The framework is therefore relational by construction: its primitive object is a pairwise comparison, not an isolated score.

3.2 Views in BSM

Scalar view. We use two scalar quantities already present in the archived analyses. The first is accuracy on disambiguated BBQ questions, reported over nine dimensions in Figure 2. The second is the benchmark-native bias score in ambiguous and disambiguated settings, reported as `s_AMB` and `s_DIS` in Table 1. Negative values indicate anti-stereotypical behavior, positive values indicate stereotypical behavior, and values near zero indicate neutrality.

Distributional view. A scalar score can hide *how* a model arrived there. BSM therefore stores full answer histograms over social groups and refusal tokens. The released figures visualize this directly for the BBQ gender dimension, while the notebooks additionally compute cosine distances and Jensen–Shannon divergences over normalized histograms for BBQ and UnQover. This view is especially informative when two models share similar bias scores but have very different refusal profiles.

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Behavioral view. For base–tuned comparisons, BSM tracks flip behavior on ambiguous examples. The most important event is a change from a contentful answer to `Unknown`. This event is not automatically a fairness improvement: in ambiguous contexts it may be appropriate, but high rates can also indicate that tuning suppresses responses without changing underlying directional preferences. We therefore interpret flip statistics jointly with forced-choice results from UnQover and with disambiguated BBQ behavior.

Representational view. When weights are available, BSM computes CKA between base and tuned hidden states on matched prompts. In the archived code, CKA is evaluated on five open-weight families with up to 64 prompts per dimension over gender, religion, nationality, and race/ethnicity. This view tests whether output-level changes coincide with substantial internal reorganization.

3.3 What BSM Is and Is Not

BSM is a framework, not a single scalar leaderboard. In principle, one could collapse the view-specific similarities into a global index, but doing so would obscure the main operational distinction we want to preserve: fairness by abstention, fairness by directional change, and fairness by representational change are not interchangeable. In this paper we therefore report BSM as a coordinated audit space, using figures and tables to show where different views agree and where they diverge.

4 Experimental Setup

4.1 Models

The archived experiments cover contemporary LLaMA and Gemma families, Gemini and GPT endpoints, GPT-2 as a legacy baseline, and older instruction-tuned systems such as Vicuna and Alpaca. Model provenance for the open-weight families follows the LLaMA 2, LLaMA 3, LLaMA 3.2, Gemma, Gemma 2, and Gemma 3 releases (Touvron et al., 2023; Dubey et al., 2024; Meta AI, 2024; Team et al., 2024a;b; 2025). Because artifact coverage differs by analysis, the disambiguated-accuracy sweep contains 30 endpoints, while the four-dimension paired BBQ audit uses the 29 archived prediction files available for that subset.

4.2 Benchmarks and Splits

The main quantitative analysis uses BBQ (Parrish et al., 2022). The released accuracy figure spans nine dimensions: age, disability, gender, nationality, physical appearance, race/ethnicity, religion, socioeconomic status, and sexual orientation. Across these dimensions, each model produces 29,372 answers, so the 30-endpoint accuracy sweep covers roughly 881,160 BBQ responses.

The pairwise audit focuses on the four shared dimensions used throughout the archived bias-score and flip analyses: gender, nationality, religion, and ethnicity. These four splits contain 8,416 ambiguous and 8,416 disambiguated examples in total. We analyze ambiguous and disambiguated contexts separately throughout the paper.

The repository also contains notebook pipelines for UnQover (Li et al., 2020) and open-ended StereoSet (Nadeem et al., 2021). The released main-paper figures and tables are concentrated on BBQ and CKA, while the UnQover and StereoSet code serves as supporting evidence for the broader BSM methodology.

4.3 Implementation of the Audit

The archived notebooks provide the operational definitions used in this paper. For BBQ, they compute disambiguated accuracy, ambiguous/disambiguated bias scores, output histograms, and flip summaries for base–tuned model pairs. For UnQover, they compute output histograms and pairwise cosine and Jensen–Shannon distances over forced-choice distributions. For CKA, they compare hidden states between base and tuned open-weight models and summarize the resulting matrices by diagonal and full-matrix means.

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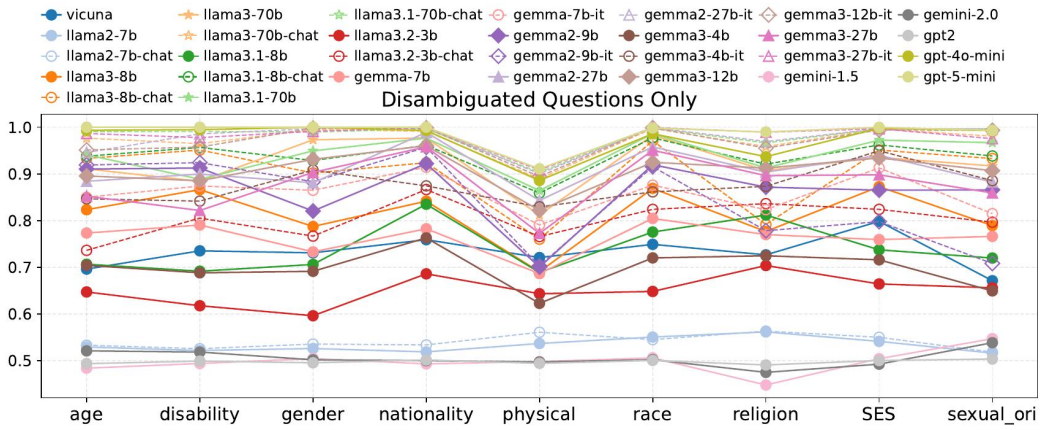

Figure 2: **Accuracy of LLMs on disambiguated BBQ questions.** Accuracy is reported across the nine archived BBQ dimensions. The strongest tuned open-weight models cluster with GPT-4o Mini and GPT-5 Mini, while Gemini 1.5/2.0 and GPT-2 remain near chance on this multiple-choice evaluation.

This setup reflects the central methodological aim of BSM: combine what can be measured from outputs for *all* models with a representation-level check available only for open-weight families. The black-box views are primary; the white-box view is diagnostic.

5 Results

5.1 Disambiguated Accuracy and Scalar Fairness Do Not Tell the Same Story

Figure 2 reports disambiguated BBQ accuracy. GPT-5 Mini achieves the highest mean accuracy across the nine archived dimensions (0.9879), followed closely by LLaMA 3.1 70B-Instruct (0.9809), Gemma 3 27B-It (0.9804), Gemma 2 27B-It (0.9767), and GPT-4o Mini (0.9756). Several open-weight tuned models therefore operate in the same high-accuracy regime as the proprietary systems in this benchmark.

At the same time, Table 1 shows that similar accuracy does not imply similar fairness signatures. LLaMA 3.1 8B exhibits one of the largest scalar shifts after tuning, dropping from 18.59 to 1.38 in `s_AMB` and from 31.37 to 4.78 in `s_DIS`. By contrast, LLaMA 3.2 3B moves in the opposite direction: tuning raises `s_AMB` from 11.95 to 15.71 and `s_DIS` from 17.67 to 30.97, indicating a fairness regression under answer-required conditions. The same table also shows that tuned Gemma 3 12B and 27B models become nearly neutral and approach the proprietary models on the reported bias scores.

5.2 Instruction Tuning Often Acts Through Abstention

The bias-flip analysis clarifies *how* many of these scalar improvements occur. Using the archived BBQ predictions and the public BBQ splits, we recomputed ambiguous-case transitions for all available base–tuned pairs over gender, nationality, religion, and ethnicity. The strongest pattern is not always a direct shift from stereotypical to anti-stereotypical answers; it is often a shift from stereotypical to `Unknown`.

This effect is especially strong in the Gemma and LLaMA 3.1 families. Among ambiguous examples where the base model selected a stereotypical answer, instruction tuning redirects 95.9% of such decisions to `Unknown` for Gemma 2 9B, 84.5% for Gemma 3 12B, 72.0% for LLaMA 3.1 70B, and 65.0% for LLaMA 3.1 8B. LLaMA 2 7B is qualitatively different: only 9.9% of its stereotypical ambiguous answers are converted to `Unknown`. Figure 3 visualizes this tendency.

This result is not merely anecdotal. Aggregated over all ambiguous examples in the four-dimension audit, the highest rates of non-`Unknown` to `Unknown` transitions occur for Gemma 2 9B→It and

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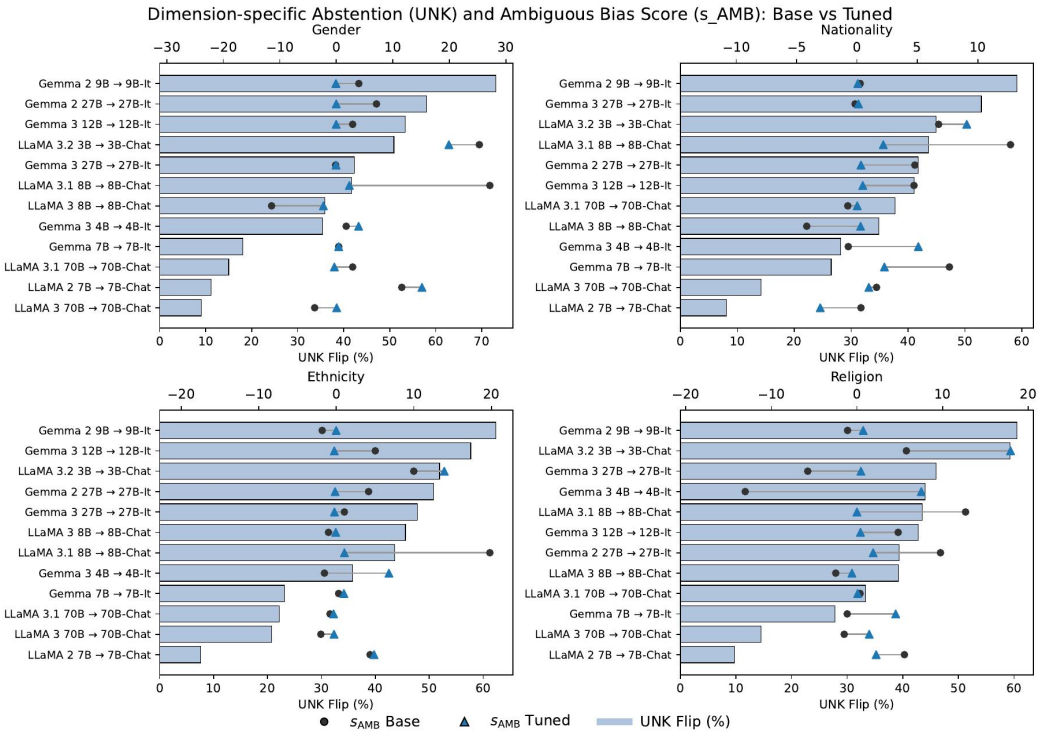

Figure 3: **Bias-flip behavior on ambiguous BBQ examples.** The archived flip analysis shows that many tuned models reduce visible bias mainly by routing ambiguous cases to `Unknown`. Recomputed from the released prediction files, the share of *previously stereotypical* ambiguous answers flipped to `Unknown` is 95.9% for Gemma 2 9B→It, 84.5% for Gemma 3 12B→It, 65.0% for LLaMA 3.1 8B→Instruct, and 9.9% for LLaMA 2 7B→Chat.

LLaMA 3.1 8B→Instruct (both 0.20 of all ambiguous items), followed by Gemma 3 12B→It (0.18). BSM therefore distinguishes two superficially similar outcomes: a model can move toward neutrality because it changes directional preferences, or because it refuses more often. The scalar view alone cannot tell these apart.

5.3 Histogram Views Reveal Family-Level Similarity

Figure 4 illustrates the histogram view for the BBQ gender dimension. The principal visual structure is family-level: base and tuned versions of the same family remain recognizably similar, but instruction tuning often increases the mass on `Unknown`. This is most visible for LLaMA 3.1, Gemma 2, Gemma 3, GPT-4o Mini, and GPT-5 Mini.

These histogram patterns also explain why BBQ and UnQover should be read together. In ambiguous BBQ settings, refusal can mask directional bias. UnQover removes that escape hatch and therefore tests whether the same models still show stereotypical leanings when forced to choose. The notebooks consistently use this complementarity: BBQ is the caution-sensitive view, UnQover is the forced-choice stress test.

5.4 Representations Change Less Than Outputs

The representational view in Table 2 supports the same interpretation. Across the five archived open-weight families, average diagonal CKA ranges from 0.941 to 0.991, and the mean over the full CKA matrix ranges from 0.851 to 0.911. In other words, base and tuned models remain highly aligned in hidden-state geometry even when their output behavior changes substantially.

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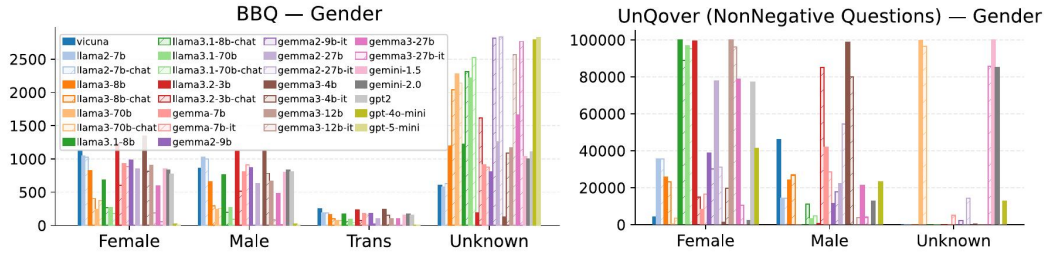

Figure 4: **Output distributions in the gender dimension.** The distributional view exposes behavioral similarity that scalar scores can hide. Several tuned models become more neutral largely by concentrating probability mass on `Unknown`, while GPT-4o Mini, GPT-5 Mini, and tuned Gemma 2/3 models combine strong neutrality with highly refusal-skewed ambiguous outputs.

Table 3: **Overall evaluation summary by model.** Qualitative synthesis of accuracy, abstention, bias direction, and representational similarity trends.

Model (abbrev.)	Key observations
Vicuna / Alpaca	Strongly anti-stereotypical; low accuracy; low abstention.
LLaMA 2 7B	Remain stereotypical after tuning; very low accuracy; weaker fairness.
LLaMA 3 8B	Anti-stereotypical lean; accuracy improves with tuning; moderate drift from LLaMA 2.
LLaMA 3.1 8B	Large bias drop after tuning; accuracy improves with tuning; high abstention.
LLaMA 3.1 70B	Near-neutral after tuning; high accuracy; high abstention even under forced choice.
LLaMA 3.2 3B	Strongly stereotypical; low accuracy; low abstention; weaker fairness vs 3.1 peers.
Gemma 2 9B	Stereotypical in base; abstention increases with tuning.
Gemma 3 4B	Slight stereotypical bias; accuracy competitive with LLaMA mid-size.
Gemma 3 12B / 27B	Near-neutral after tuning; high CKA similarity; fairness competitive with closed ones.
Gemini 1.5 / 2.0	Strong abstention; 2.0 skews anti-stereotypical; very low accuracy.
GPT-2	Extremely stereotypical bias; very low accuracy and fairness; serves as legacy baseline.
GPT-4o Mini	Near-zero bias; high accuracy; balanced abstention–fairness.
GPT-5 Mini	Near-perfect neutrality; highest accuracy; strongest stability across metrics.

This does not imply that tuning is ineffective. It implies something more specific and operationally important: much of the observed fairness change in the archived runs appears to be realized through output-layer or policy-level behavior rather than wholesale representational reorganization. BSM makes this visible, because the behavioral and representational views can be compared directly.

Table 1: **Average bias scores.** "−": anti-stereotypical, "+": stereotypical, and values near 0 = neutrality. Shown for ambiguous (`s_AMB`) and disambiguated (`s_DIS`) contexts.

Base Model	Avg. `s_AMB`		Avg. `s_DIS`	
	Base	Tuned	Base	Tuned
LLaMA 2 7B	5.45	4.30	7.50	6.65
LLaMA 3 8B	-4.78	-0.66	-8.72	-2.10
LLaMA 3 70B	-1.42	0.55	-4.51	2.50
LLaMA 3.1 8B	18.59	1.38	31.37	4.78
LLaMA 3.1 70B	0.42	-0.15	0.81	-1.44
LLaMA 3.2 3B	11.95	15.71	17.67	30.97
Gemma 7B	1.81	2.05	0.69	0.95
Gemma 2 9B	0.08	0.18	6.83	-2.02
Gemma 2 27B	6.95	0.51	14.31	-1.45
Gemma 3 4B	-3.89	5.83	2.69	8.62
Gemma 3 12B	4.36	0.15	6.12	-0.17
Gemma 3 27B	-1.25	0.07	-0.26	-1.50
Gemini 1.5	–	2.37	–	3.07
Gemini 2.0	–	-4.17	–	-5.54
GPT-2	72.43	–	96.19	–
GPT-4o Mini	–	0.47	–	2.66
GPT-5 Mini	–	0.21	–	1.10

6 ANALYSIS

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The released artifacts support three main conclusions about comparative fairness.

Table 2: Average CKA scores.

Model	Diag	Full
LLaMA 2 7B	0.991	0.902
LLaMA 3 8B	0.973	0.851
Gemma 1 7B	0.981	0.896
Gemma 2 9B	0.941	0.906
Gemma 3 12B	0.972	0.911

Fairness gains are often policy gains. The most consistent pattern in the archive is that instruction tuning suppresses harmful visible behavior by increasing abstention on ambiguous prompts. This can be desirable in deployments that prefer caution under uncertainty. But BSM shows why it should not be misread as equivalent to deep debiasing. The combination of strong `Unknown` flips, refusal-heavy histograms, and high CKA indicates that some tuned models are changing *decision policy* more than *internal geometry*.

Forced-choice settings remain necessary. A model that looks neutral in BBQ ambiguous contexts may still reveal stereotypical preferences when abstention is removed. This is exactly why BSM combines BBQ with UnQover rather than selecting one benchmark as the canonical answer. In procurement or regression settings, the practical question is often whether a system stays safe when it must answer, not only when it is allowed to refuse.

Scaling and tuning are not monotonic with fairness. The archive contains both improvement and regression stories. LLaMA 3.1 8B improves sharply after tuning on the scalar view, while LLaMA 3.2 3B becomes more stereotypical in `s_DIS`. GPT-5 Mini is both highly accurate and close to neutral on the reported BBQ scalar metrics, but tuned Gemma 3 12B and 27B are already competitive on those same bias scores. BSM therefore discourages simplistic assumptions such as "newer is fairer," "bigger is fairer," or "closed is fairer."

View disagreement is operationally informative. BSM is most useful when its views disagree. Small scalar differences with large histogram distances indicate that two models reach similar headline scores through different answer policies. Neutrality on ambiguous BBQ paired with weak answer-required behavior points to caution without robust stereotype resolution. Large ambiguous-case flip rates alongside near-identity CKA suggest that tuning mainly changes output policy rather than the underlying representation. In practice, these disagreements help separate models that are suitable for high-abstention assistance settings from models that remain risky in forced-decision workflows.

The views also separate deployment profiles. The archived models do not collapse to a single fairness ordering once BSM is applied. LLaMA 3.1 8B-Instruct is a caution-oriented profile: scalar bias drops sharply, but much of that gain is mediated by ambiguous-case routing to `Unknown`. Gemma 3 27B-It is a different case, combining near-neutral `s_AMB`/`s_DIS` with high CKA to its base model, which is more consistent with a narrower behavioral correction. LLaMA 3.2 3B-Instruct remains the clearest regression case, especially in answer-required settings. GPT-5 Mini instead serves as the strongest archive reference point because it pairs the best disambiguated accuracy with near-zero scalar bias.

These findings motivate concrete audit workflows. For *procurement*, pairwise similarity can identify open-weight substitutes whose fairness signatures are close to a target proprietary model under the relevant prompt regime. For *release regression*, base–tuned flip analysis can distinguish genuine directional improvement from a refusal-only change. For *lineage screening*, histogram and cosine views can reveal when a new release remains behaviorally close to an older biased family despite cosmetic scalar gains.

7 Conclusion

This paper argued that fairness should be audited relationally. BSM reframes fairness evaluation as the comparison of multi-view bias signatures, combining scalar outcomes, output distributions, behavioral flips, and representational similarity. Applied to the released LLM audit archive, BSM reveals that instruction tuning often reduces benchmark-visible bias by increasing abstention on ambiguous cases, while internal geometry stays highly stable. It also shows that tuning does not monotonically improve fairness: some models become substantially more neutral, while others regress under answer-required settings.

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The broader lesson is methodological. Isolated benchmark scores are too coarse for practical model governance. Comparative audits make it possible to ask whether two models fail similarly, whether a release meaningfully changes behavior, and whether apparent fairness improvements reflect caution or actual directional change. BSM is designed for exactly those questions.

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